LAKSHYA

JEE 2027

Lecture No.- 23

MATHEMATICS

RELATIONS & FUNCTIONS

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Topics *to be covered*

1. P&C Based Problems
2. Composite Function
3. Question Practice
4.

LAST CLASS REVISION

Answer the following Questions as True/False:

Notes

1. If $\boxed{f: R \to R}$ and $\boxed{f(x) \text{ is continuous function}}$ such that as $x \to \infty \Rightarrow f(x) \to -\infty$ and $x \to -\infty \Rightarrow f(x) \to +\infty$ then $f(x)$ is an onto function.

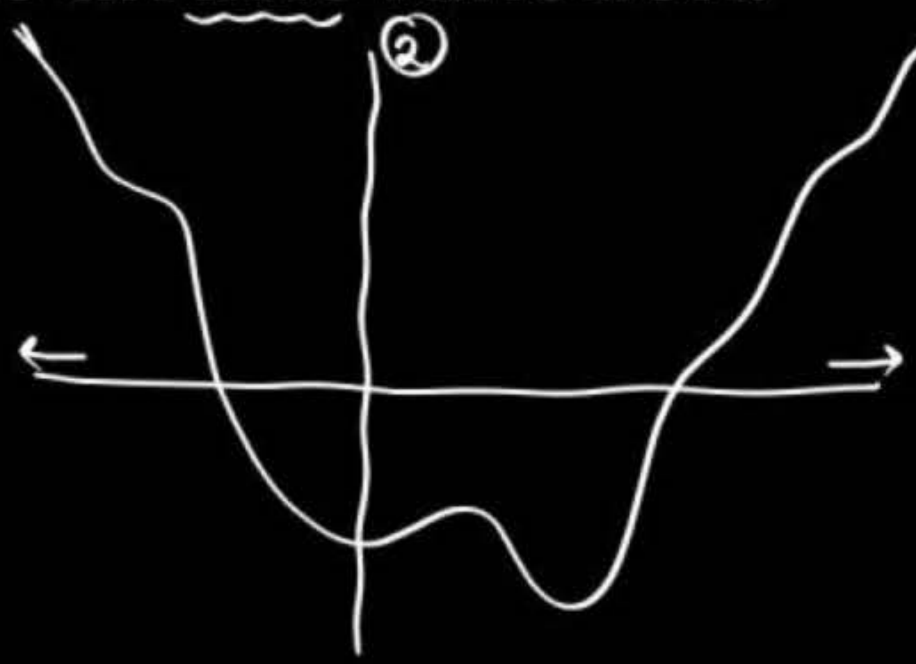
T

2. If $f: R \to R$ and $f(x)$ is continuous function such that as $\boxed{x \to \pm\infty} \Rightarrow \boxed{f(x) \to +\infty}$ then $f(x)$ is an Into function.

T

1

2



Question Practice

Bumper Question

Using P & C and proper case making answer following questions:

(i) In $n(A) = 5$, then number of $\boxed{\text{injections}}$ $f: A \to A$ such that $f(i) \neq i$ is 44.

1-1

A: 1, 2, 3, 4, 5

A: 1, 2, 3, 4, 5

derang. of 5 things

* X!

$f(1) \neq 1$

$f(2) \neq 2$

$\vdots$

$f(5) \neq 5$

$$D_n = n!\left(\frac{1}{0!} - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} \cdots + \frac{(-1)^n}{n!}\right)$$

$$D_5 = 5!\left(\cancel{\frac{1}{0!} - \frac{1}{1!}} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!}\right)$$

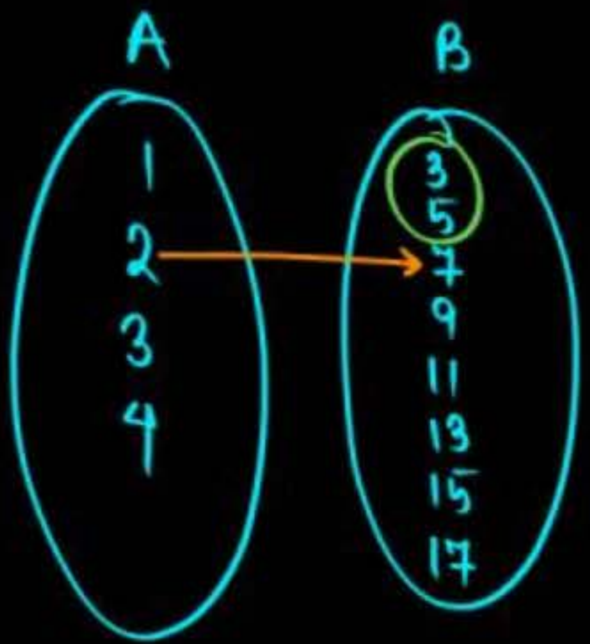
$$= 60 - 20 + 5 - 1$$

$$= 44$$

Bumper Question

Using P & C and proper case making answer following questions:

(ii) If $f : A \to B, A = \{1, 2, 3, 4\}, B = \{3, 5, 7, 9, 11, 13, 15, 17\}$ and $f(2) = 7$ then (number of strictly increasing function) + (number of strictly decreasing function f) is 25.



$$\boxed{f(1) < f(2) < f(3) < f(4)}$$

$$2 \times 1 \times {}^5C_2 = 20$$

$$f(1) > \overset{7}{f(2)} > f(3) > f(4)$$

$$^5C_1 \times 1 \times 1 = 5$$

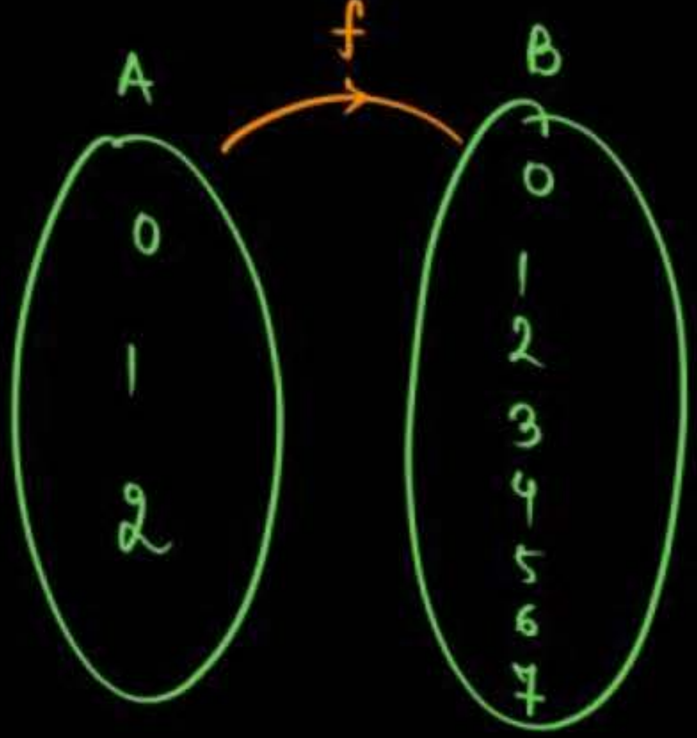
Bumper Question

Using P & C and proper case making answer following questions:

(iii) If $f:\{0,1,2\} \to \{0,1,2,3,\dots.7\}$ then total number of functions satisfying $\boxed{f(i) \le f(j)}\ \forall\, i < j$ in domain.

Increasing

$\boxed{f(0) \le f(1) \le f(2)}$



$\boxed{f(0) = f(1) = f(2)} \longrightarrow 8$

$f(0) < f(1) < f(2) \longrightarrow {}^8C_3 \times 1 = 56$

$f(0) < f(1) = f(2) \longrightarrow {}^8C_2 \times 1 = 28$

$f(0) = f(1) < f(2) \longrightarrow {}^8C_2 \times 1 = 28$

120 func^n.

$$\boxed{\begin{array}{l} \forall\, i<j \Rightarrow f(i) \le f(j) \\ \forall\, i<j \Rightarrow f(i) < f(j) \end{array}}$$

Increasing

St. Increasing

Question [JEE Mains-2024]

79%

Let $[t]$ be the greatest integer less than or equal to t. Let A be the set of all prime factors of 2310 and $f: A \to \mathbb{Z}$ be the function $f(x) = \left[\log_2\left(x^2 + \left[\frac{x^3}{5}\right]\right)\right]$. The number of one-to-one functions from A to $\boxed{\text{the range of } f}$ is

(A) 20

(B) 120

(C) 25

(D) 24

$2310 = \boxed{2 \times 3 \times 5 \times 7 \times 11}$

$f(2) = [\log_2(5)] = 2$

$f(3) = [\log_2 14] = 3$

$f(5) = [\log_2(50)] = 5$

$f(7) = 6$

$f(11) = 8$

A	Range.
2	2
3	3
5	5
7	6
11	8

$1\text{-}1 = 5!$

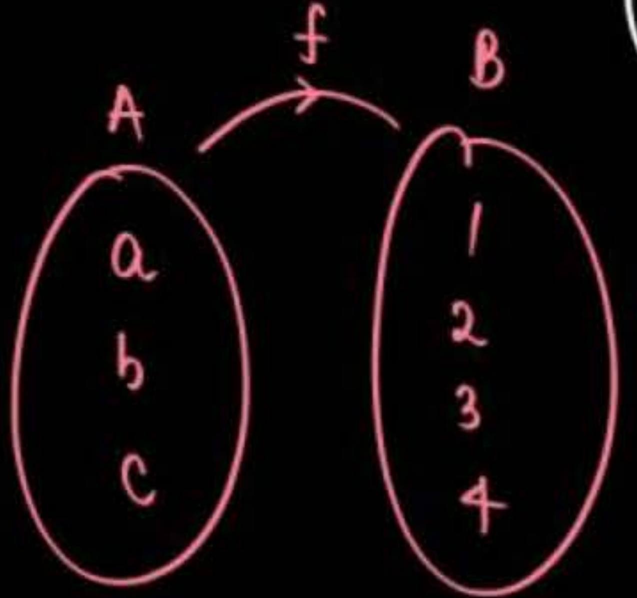
$= 120$

Critical Thinking Question [JEE Mains-2020]

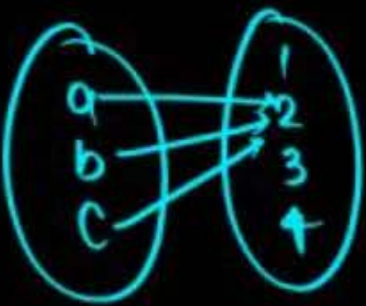
Let $A=\{a,b,c\}$ and $B=\{1,2,3,4\}$. Then the number of elements in the set $C=\{f : A\to B \mid 2\in f(A)$ and f is not one-one$\}$ is ___ 19 funcⁿ

Range.

$\boxed{\text{2 Range में होगा}}$ & $\boxed{\text{Many-one.}}$

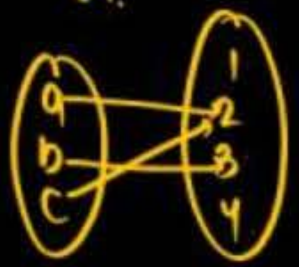


(I) $\boxed{\text{all 3 mapped with 2}}$ → 1



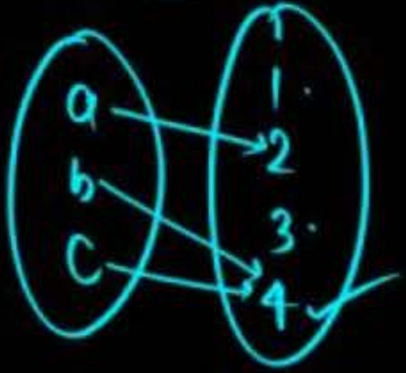
(II) $\boxed{\text{exactly two are mapped 2}}$ → $^3C_2\times 3 = 9$

ex:-



(III) exactly one is mapped with 2

ex:-



$^3C_1\times{}^3C_1\times 1 = 9$

Challenger Question [JEE Mains-2022]

Case-Making

The number of one-one functions $f:\{a,b,c,d\}\to\{0,1,2,\ldots,10\}$ such that $2f(a)-f(b)+3f(c)+f(d)=0$ is

$2f(a)+3f(c)+f(d)=f(b)$

2×2	3×1	0	7
2×3	3×1	0	9
2×1	3×2	0	8
2×2	3×0	1	5
2×3	3×0	1	7
2×4	3×0	1	9
2×0	3×2	1	7

H.W. — A

Ans.

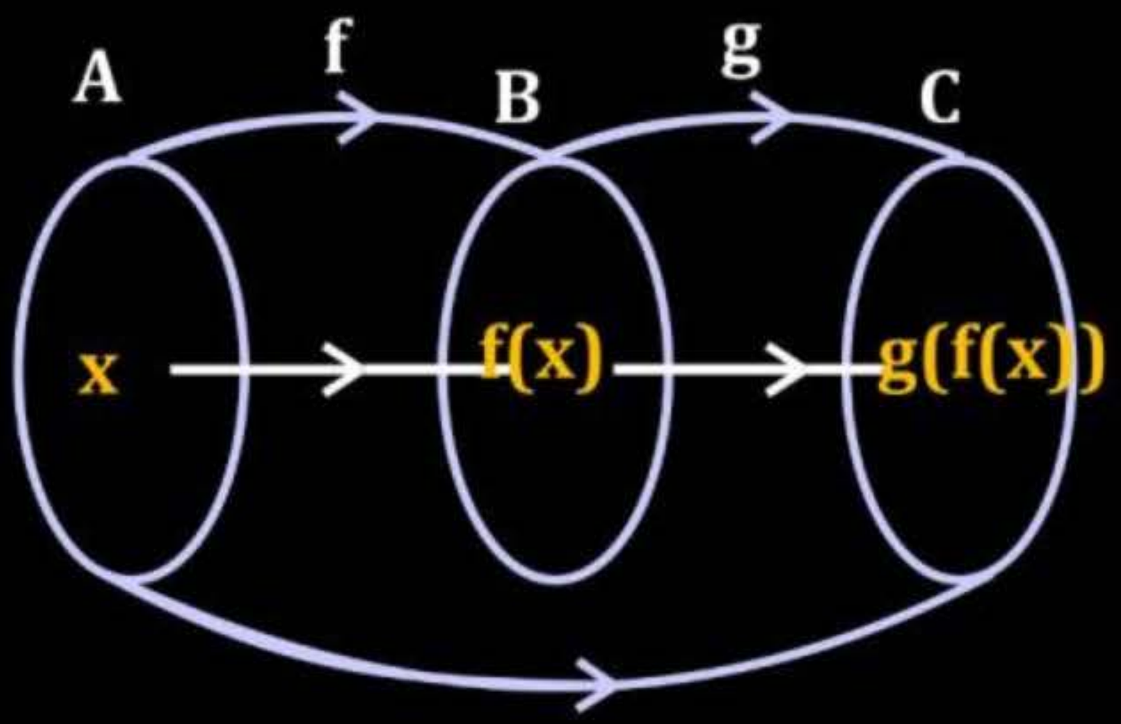
Critical Thinking Question [JEE Mains-2021]

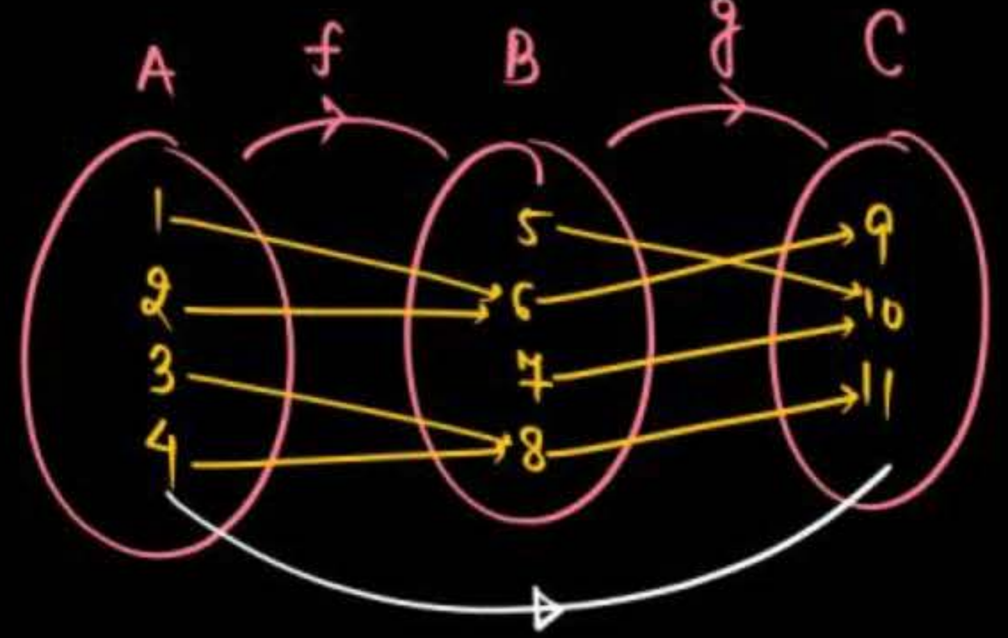
Let $S = \{1, 2, 3, 4, 5, 6, 7\}$. Then the number of possible function $f : S \rightarrow S$ such that $f(m \cdot n) = f(\mathrm{m}) \cdot f(n)$ for every $m, n \in S$ and $m \cdot n \in S$ is equal to

H.W.

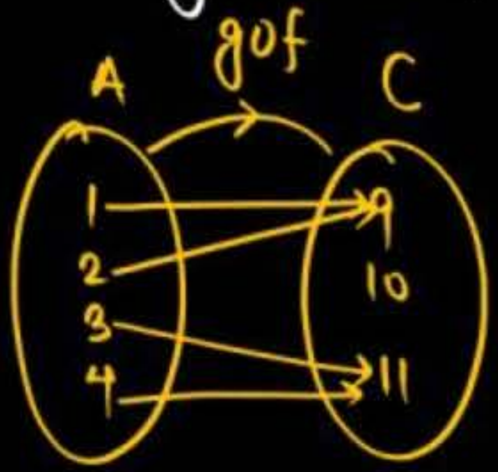
COMPOSITE FUNCTIONS

"Function inside function"





$g \circ f = g(f(x))$



IMPORTANT POINTS:

Note :

1. In general : $\boxed{gof(x) \neq fog(x)}$
2. For composite function $g(f(x))$ to be defined range of $f(x)$ is subset of domain of $g(x)$.
3. Composition of two bijective function is also **Bijective.**
4. If domain of $y = f(x)$ is $x \in [\alpha, \beta)$ then domain of $y = f(g(x))$ is solution set of $\boxed{\alpha \leq g(x) < \beta}$ or $g(x) \in [\alpha, \beta)$.

$\overbrace{\underbrace{f}_{\text{Bijective}}(\underbrace{g}_{\text{Bijective}}(x))}^{\text{Give}} \Rightarrow$ Bijective.

Acidic Question

Answer the forewing questions:

(i) If $f(x)=\frac{x}{1-x}$, then domain of $f(f(f(x)))$ is ______. $x \in R - \left\{1, \frac{1}{2}, \frac{1}{3}\right\}$

M-1

$x = f(x)$

$$f(f(x)) = \frac{f(x)}{1-f(x)} = \frac{\left(\frac{x}{1-x}\right)}{1-\left(\frac{x}{1-x}\right)}$$

$x = f(x)$

$$f(f(f(x))) = \frac{\left(\frac{f(x)}{1-f(x)}\right)}{1-\left(\frac{f(x)}{1-f(x)}\right)} = \frac{\frac{\left(\frac{x}{1-x}\right)}{1-\left(\frac{x}{1-x}\right)}}{1-\frac{\left(\frac{x}{1-x}\right)}{1-\left(\frac{x}{1-x}\right)}}$$

$1-x \neq 0 \rightarrow x \neq 1$

$1-\frac{x}{1-x} \neq 0 \rightarrow x \neq \frac{1}{2}$

$1-\frac{\left(\frac{x}{1-x}\right)}{1-\left(\frac{x}{1-x}\right)} \neq 0 \rightarrow x \neq \frac{1}{3}$

Acidic Question

Answer the forewing questions:

(i) If $f(x) = \frac{x}{1-x}$, then domain of $f(f(f(x)))$ is ______. $x \in R - \left\{1, \frac{1}{2}, \frac{1}{3}\right\}$

domain$(x) \in R - \{1\}$ ✓

$x = f(x)$

$f(f(x)) = \frac{f(x)}{1-f(x)}$, $f(x) \neq 1$ OR $\frac{x}{1-x} \neq 1 \Rightarrow x \neq \frac{1}{2}$

$x = f(x)$

$f(f(f(x))) = \frac{f(f(x))}{1-f(f(x))}$, $f(f(x)) \neq 1 \Rightarrow \frac{f(x)}{1-f(x)} \neq 1 \Rightarrow f(x) \neq \frac{1}{2} \Rightarrow \frac{x}{1-x} \neq \frac{1}{2} \Rightarrow x \neq \frac{1}{3}$

Acidic Question

Answer the forewing questions:

(i) If $\boxed{f(x)=\frac{x}{1-x}}$, then domain of $f(f(f(x)))$ is ______. $x \in R - \left\{1, \frac{1}{2}, \frac{1}{3}\right\}$

M-1

$x = f(x)$

$$f(f(x)) = \frac{f(x)}{1-f(x)} = \frac{\left(\frac{x}{1-x}\right)}{1-\left(\frac{x}{1-x}\right)} = \frac{x}{\cancel{(1-x)}\left(\frac{1-x-x}{\cancel{1-x}}\right)} = \boxed{\frac{x}{1-2x}}$$

$x = f(x)$

$$f(f(f(x))) = \frac{f(x)}{1-2f(x)} = \frac{x}{(1-x)\left(1-\frac{2x}{1-x}\right)} = \boxed{\frac{x}{1-3x}}$$

* $$\underbrace{f(f(f(\cdots}_{47\text{ Times}} f(100)))\ldots) = \frac{100}{1-47\times 100}$$

$$\frac{x}{1-47x}$$

Acidic Question

Answer the forewing questions:

(ii) If $f(x)=\frac{1}{x}$, $h(x)=\frac{1}{4x^2-1}$ & $g(x)=\frac{5x}{x+2}$, then domain of $gohof(x)$ is ___.

$x \neq -2$

$x \in R - \left\{\pm\frac{1}{2}, \pm 2, 0, \pm 2\sqrt{2}\right\}$

$\boxed{x \neq 0}$ $\qquad$ $\boxed{x \neq \pm\frac{1}{2}}$

$x = f(x)$

$x = h(f(x))$

$g(h(f(x)))$

$$g(h(f(x))) = \frac{5h(f(x))}{h(f(x))+2}, \quad h(f(x)) \neq -2$$

$$h(f(x)) = \frac{1}{4f^2(x)-1}, \quad f(x) \neq \pm\frac{1}{2} \Rightarrow \frac{1}{x} \neq \pm\frac{1}{2} \Rightarrow \boxed{x \neq \pm 2}$$

$\frac{1}{4f^2(x)-1} \neq -2$

$-\frac{1}{2} \neq 4f^2(x)-1$

$\frac{1}{2} \neq 4f^2(x)$

$\frac{1}{8} \neq f^2(x)$

$\pm\frac{1}{2\sqrt{2}} \neq f(x)$

$\pm\frac{1}{2\sqrt{2}} \neq \frac{1}{x}$

$\boxed{x \neq \pm 2\sqrt{2}}$

Acidic Question

Answer the forewing questions:

$x = f(x) \Rightarrow f(f(x)) = \frac{1}{1-f(x)} = \frac{1}{1-\frac{1}{1-x}}$

$= \frac{1-x}{1-x-1} = \frac{1-x}{-x} = \frac{x-1}{x}$

(iii) Let $f(x) = \frac{1}{1-x}, x \in R - \{0, 1\}$ and $f(f(x)) = f^2(x)$, $f(f(f(x))) = f^3(x)$, so on, then $f^{2022}(2023) =$ 2023 ________.

Ans

$f(f(x)) = \frac{x-1}{x}$

$x = f(x)$

$f^{2022}(x) = \underbrace{f(f(f \dots f}_{2022 \text{ Times}}(x)) \dots)$

$f(f(f(x))) = \frac{f(x)-1}{f(x)} = \frac{\frac{1}{1-x} - 1}{\frac{1}{1-x}} = x$

$f^{2022}(x) = x$

$x = f(x)$

$f(\overbrace{f(f(f(x)))}^{x}) = f(x)$

Question [JEE Mains-2022]

$f(x) = \left((2-x^{25})(2+x^{25})\right)^{\frac{1}{50}} = (4-x^{50})^{\frac{1}{50}}$

Let $f: R \to R$ be a function defined by $f(x) = \left(2\left(1-\frac{x^{25}}{2}\right)(2+x^{25})\right)^{\frac{1}{50}}$

If the function $g(x) = f(f(f(x))) + f(f(x))$, the greatest integer less than or equal to $g(1)$ is ______.

$\Rightarrow f(f(x)) = \left(4 - \underbrace{(f(x))^{50}}\right)^{\frac{1}{50}}$

$= \left(\underbrace{\cancel{4} - (\cancel{4} - x^{50})}\right)^{\frac{1}{50}} = x$

$(x^{50})^{\frac{1}{50}}$

$g(x) = f(x) + x$

$x = 1$

$[g(1)] = [f(1) + 1] = [3^{\frac{1}{50}} + 1] = 2$ Ans

$3^{\frac{1}{50}} \to 1.\ldots$

Question [JEE Mains-2022]

Let $f(x)$ and $g(x)$ be two real polynomials of degree 2 and 1 respectively. If $f(g(x)) = 8x^2 - 2x$, and $g(f(x)) = 4x^2 + 6x + 1$, then the value of $f(2) + g(2)$ is ______.

HW.

COMPOSITE OF NON-UNIFORMLY DEFINED FUNCTIONS

To define $\boxed{f(g(x))}$:

S-(I) Put "$g(x)$" in $f(x)$ wherever 'x' is present.

S-(II) Draw graph of inside function (i.e. $g(x)$).

S-(III) "See graph of $g(x)$", remove $g(x)$ by expression & interval of 'x'.

Question

Given $f(x) = \begin{cases} 1-x, & \text{if } x \le 0 \\ x^2, & \text{if } x > 0 \end{cases}$ and $g(x) = \begin{cases} -x, & \text{if } x < 1 \\ 1-x, & \text{if } x \ge 1 \end{cases}$, find fog and gof.

$fog \rightarrow f(g(x))$

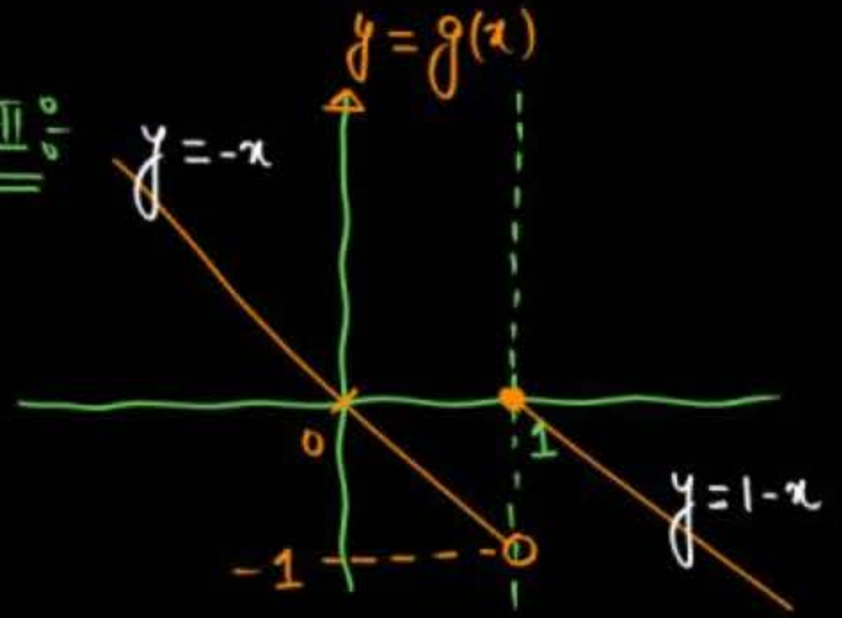
$gof \rightarrow$ HW.

$x = g(x)$

S-I:-

$$\boxed{\begin{aligned} f(g(x)) &= 1-g(x), & g(x) \le 0 \\ &= (g(x))^2, & g(x) > 0 \end{aligned}}$$

S-II:-



S-III:-

$$\boxed{\begin{aligned} g(x) \le 0 &\Rightarrow x \in [0,1) \cup [1,\infty) \\ g(x) > 0 &\Rightarrow x \in (-\infty, 0) \end{aligned}}$$

*

S-IV:-

$$\begin{aligned} f(g(x)) &= 1-g(x), & x \in [0,1) \\ &= 1-g(x), & x \in [1,\infty) \\ &= (g(x))^2, & x \in (-\infty, 0) \end{aligned}$$

S-V:

$$f(g(x)) = 1-(-x), \quad x \in [0,1)$$
$$= 1-(1-x), \quad x \in [1,\infty)$$
$$= (-x)^2, \quad x \in (-\infty,0)$$

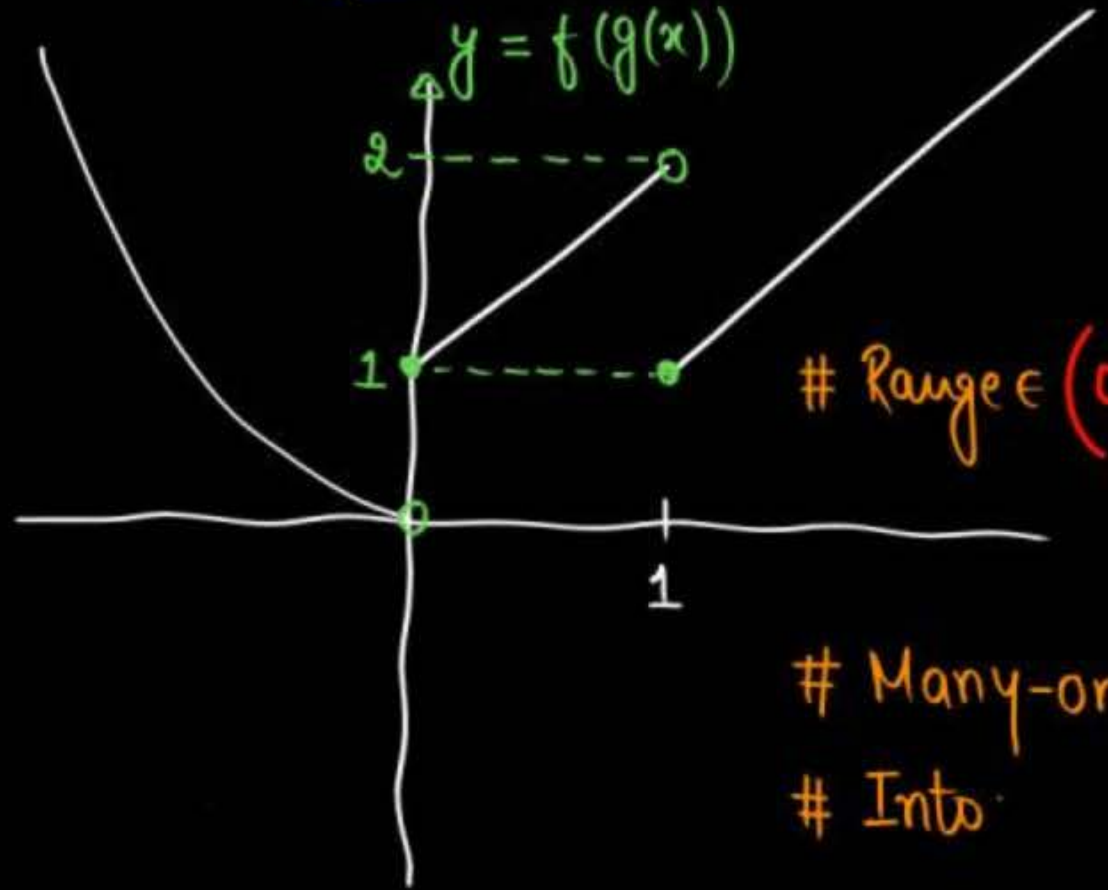
OR

$$\boxed{\begin{aligned} f(g(x)) &= x^2, & x &\in (-\infty,0) \\ &= 1+x, & x &\in [0,1) \\ &= x, & x &\in [1,\infty) \end{aligned}}$$

$f(g(3)) = 3$.

$fog : R \to R$

Graph



Range $\in (0,\infty)$

Many-one

Into

Question [JEE Mains-2024]

If $f(x) = \begin{cases} 2+2x, -1 \leq x < 0 \\ 1-\frac{x}{3}, 0 \leq x \leq 3 \end{cases}$; $g(x) = \begin{cases} -x, -3 \leq x \leq 0 \\ x, 0 < x \leq 1 \end{cases}$, then range of $(f o g(x))$ is

HW.

(A) (0, 1]

(B) [0, 3)

(C) [0, 1]

(D) [0, 1)

DO IT BY YOURSELF (DIBY)

DIBY-154

Let $A = \{1, 3, 7, 9, 11\}$ and $B = \{2, 4, 5, 7, 8, 10, 12\}$. Then the total number of one-one maps $f: A \to B$, such that $f(1) + f(3) = 14$ is: [JEE Mains-2024]

(A) 180 (B) 120 (C) 480 (D) 240

[Ans. : (D)]

DIBY-155

Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{1, 2, 3, 4, 5, 6\}$. Then the number of functions $f: A \to B$ satisfying $f(1) + f(2) = f(4) - 1$ is equal to [JEE Mains-2023]

[Ans. : 360]

DIBY-156

The number of bijective functions $f: \{1, 3, 5, 7, \ldots, 99\} \to \{2, 4, 6, 8, \ldots, 100\}$, such that $f(3) \geq f(9) \geq f(15) \geq f(21) \geq \cdots \geq f(99)$ .... [JEE Mains-2022]

(A) ${}^{50}P_{17}$ (B) ${}^{50}P_{33}$ (C) $33! \times 17!$ (D) $\frac{50!}{2}$

[Ans. : (C)]

DIBY-157

Let $S=\{1,2,3,4\}$. Then the number of elements in the set $\{f: S\times S\rightarrow S: f$ is onto and $f(a,b)=f(b,a)\geq a\,\forall(a,b)\in S\times S\}$ is [JEE Mains-2022]

[Ans. : 37]

DIBY-158

The total number of functions, $f:\{1,2,3,4\}\rightarrow\{1,2,3,4,5,6\}$ such that $f(1)+f(2)=f(3)$, is equal to: [JEE Mains-2022]

(A) 60 (B) 90 (C) 108 (D) 126

[Ans. : (B)]

DIBY-159

Let $A=\{1,2,3,5,8,9\}$. Then the number of possible functions $f: A\rightarrow A$ such that $f(m\cdot n)=f(m)\cdot f(n)$ for every $m,n\in A$ with $m\cdot n\in A$ is equal to [JEE Mains-2023]

[Ans. : 432]

DIBY-160

Let $R=\{a, b, c, d, e\}$ and $S=\{1,2,3,4\}$. Total number of onto function $f: R \rightarrow S$ such that $f(a) \neq 1$, is equal to [JEE Mains-2023]

[Ans. : 180]

Homework

Re-attempt all good Questions of Class

DIBY

It is never about end result
It's always about Journey

THANK

You

#futureIITians